

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with experience building production image and video AI systems across detection, tracking, classification, and real-time inference. Experienced in CNNs, Vision Transformers, OpenCV, PyTorch/TensorFlow, model optimisation, and deployment under latency, throughput, and reliability constraints. Built and validated end-to-end CV pipelines spanning data curation, annotation workflows, training, benchmarking, edge deployment, monitoring, and post-deployment failure analysis. Recent project work includes foundation-model and Vision Transformer evaluation using RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 for production-style CV benchmarking.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Improved production video inference reliability by adding structured logging, health checks, retries, and safe restart behaviour, reducing live pipeline crashes by 67% in 24/7 operation.
- Implemented image preprocessing and artefact-detection workflows with OpenCV to improve camera-feed quality before inference, reducing false positives by 13% in production vision pipelines.
- Built synthetic image generation and augmentation procedures with OpenCV and scikit-image to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, TensorFlow Model Zoo) and quantify accuracy, latency, and stability pre and post optimisation for regression detection.
- Implemented Git and Jenkins-based continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.

LiVNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50% with Keras and Intel OpenVINO.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.

- Explored semantic and instance segmentation models, including DeepLabv3+, Mask R-CNN, U-Net-style architectures, and SegFormer, to support annotation review, foreground/background separation, and failure analysis for industrial camera feeds.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an automotive claims assessment pipeline combining computer vision-based damage detection with severity classification to support faster triage of real-world vehicle damage scenarios.
- Built a multimodal image-to-structured-output workflow that converted vehicle-damage detections and severity predictions into insurer-friendly claim summaries using structured reasoning and LLM-generated explanations.
- Designed the full workflow across data preprocessing, model training, evaluation, and a user-facing inference module, demonstrating end-to-end delivery from model output to usable product insight.
- Created interpretable repair insights from CV model outputs, translating visual predictions into clear claim explanations for non-technical users.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Computer Vision & Image Processing:** Data augmentation, Image processing, Object detection, OpenCV, PCL (Point Cloud Library), Sensor calibration
- **Datasets, Annotation & Evaluation:** Cityscapes, COCO, custom industrial datasets, F1-score, ImageNet, IoU, KITTI, latency benchmarking, mAP, model-size benchmarking, PASCAL VOC, precision/recall
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, OpenCV, Pandas, PG vector, PIL, PostgreSQL, Python, Qt, scikit-image, SciPy
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering